

BOM - ESP32-S3 DRV8251A Motorshield V4.0

Designator	Quantity	Description	Footprint	Manufacturer	Manufacturer Part Number
A0, A1, A2, A3	4	Solder Jumper			
C1	1	Polymer Aluminium Capacitor Electrolyt, 47uF, 50VDC	SMD,D6.3xL8mm	Kyocera AVX	APD0609470M050R
C2	1	Polymer Aluminium Capacitor Electrolyt, 100uf, 63VDC	SMD,D8xL10.5mm	Kyocera AVX	APD0809101M050R
C3, C5, C7, C8, C10, C11, C14, C16, C17, C18, C19, C20, C21, C22, C23, C24, C25, C32, C38	19	Ceramic Capacitor MLCC, 100nF, 50VDC	0603	Kyocera AVX	06035C104KAZ4A
C4, C6, C12, C13, C37	5	Ceramic Capacitor MLCC, 10uF	0805	Murata Electronics	GRM21BZ71E106KE15L
C9, C15, C33, C34, C35, C36	6	Ceramic Capacitor MLCC, 10nF	0603	Kyocera AVX	06035C103K4T4A
C26, C27, C28, C29, C30, C31	6	Aluminium Capacitor Electrolyt, 22uF, 50VDC	SMD,D6.3xL5.8mm	Panasonic	EEH-ZC1H220P
D1	1	Diode, super barrier rectifiers	SOD-123F	Diodes Incorporated	SBR3U40S1FQ-7
D2, D3, D4, D5, D6, D7	6	TVS Diode, unidirectional ESD/Surge Protection 15V	SOD-323	Bourns	CDSOD323-T15S
IC1	1	Integrated Circuit, level shifter I2C	TSSOP-8	NXP	PCA9617ADPJ
IC2, IC6	2	Integrated Circuit, LED Drivers/LED Drivers, I2C	TSSOP-28	NXP	PCA9685PW,118
IC3, IC4	2	Integrated Circuit, Data Acquisition/Analog to Digital Converters	X2-QFN-10(1.5x2)	Texas Instruments	ADS1015IRUGR
IC5	1	Integrated Circuit, Interface I/O Expander 16 channel with interrupt	SSOP-28	Microchip	MCP23017T-E/SS
J1, J2	2	Connector, 22ckt, 1 row, pitch: 2.54mm, female	Plugin, P = 2.54mm	SAMTEC	CES-122-02-S-S
J3, J4, J6, J7, J8	5	Connector, 8ckt, 1 row, pitch: 2.54mm, female	Plugin, P = 2.54mm	Molex	90147-1108
J5	1	Connector, 2 ckt, screw terminal, pitch: 5mm, 12A, blue	Plugin, P = 2.54mm	Würth Elektronik	691101710002
J9	1	Connector, 8ckt, 1 row, pitch: 2.54mm, male, yellow	Plugin, P = 2.54mm	Würth Elektronik	61300811121
J10	1	Connector, 8ckt, 1 row, pitch: 2.54mm, male, red	Plugin, P = 2.54mm	Würth Elektronik	61300811121
J11	1	Connector, 8ckt, 1 row, pitch: 2.54mm, male, black	Plugin, P = 2.54mm	Würth Elektronik	61300811121
J13, J15, J17, J18	4	Connector, 2 ckt, screw terminal, pitch: 5mm, 12A, green	Plugin, P = 5mm	Phoenix Contact	5442170
J14, J16	2	Connector, 2 ckt, screw terminal, pitch: 5mm, 12A, black	Plugin, P = 5mm	Phoenix Contact	5441537
J19	1	Connector, 7 ckt, 1 row, pitch: 2.54mm, female	Plugin, P = 2.54mm	Molex	90147-1107
J25	1	Connector, QWILC, wire to board	SMD,P=1mm,Surface Mount, right Angle	JST (JAPAN SOLDERLESS TERMINALS)	SM04B-SRSS-TB(LF)(SN)
Q1	1	P-MOSFET, 1 channel	SO-8	Diodes Incorporated	DMP3020LSS-13
R1, R3, R6, R14, R16, R17, R44, R45, R46, R47	10	Resistor 10kohm, 1%	0603	Vishay	MCT0603MD1002DP500
R2	1	Resistor 4.7kohm, 1%	0603	YAGEO	RC0603FR-104K7L
R4	1	Resistor 2.7kohm, 1%	0603	Vishay	MCT06030C2701FP500
R5	1	Resistor 6.8kohm, 1%	0603	YAGEO	RC0603FR-136K8L
R7, R15, R48, R49, R50, R51	6	Resistor 47kohm, 1%	0603	YAGEO	AC0603FR-1047KL
R8, R10, R11, R12	4	Resistor 2.2kohm, 1%	0603	YAGEO	RC0603FR-102K2L
R9, R22, R25, R26, R27, R28	6	Resistor 1.3kohm, 1%	0603	Yageo Group	AC0603FR-071K3L
R13, R19, R20, R21, R23, R24, R29, R30, R31, R32, R33, R34, R35, R36, R37, R38, R39, R40, R41, R42	20	Resistor 22ohm, 1%	0603	KOA Speer	RK73H1JTTD22R0F
S1	1	Switch	Throughhole	E-Switch	RA1H1C112R
TP1, TP2, TP3, TP4, TP5, TP6, TP7, TP8, TP9, TP10, TP11, TP12, TP13, TP14, TP15, TP16, TP17, TP18, TP19, TP20, TP21, TP22, TP23, TP24, TP25, TP26, TP27, TP28, TP29, TP30, TP31, TP32, TP33, TP34, TP35, TP36, TP37, TP38, TP39, TP40, TP41, TP42, TP43, TP44, TP45, TP46, TP47, TP48, TP49, TP50, TP51	51	Testpoints	Pad round Ø: 1.5mm		
U1	1	Power Supply, DC DC CONVERTER 5V 15W	Throughhole	Traco Power	TSR 3-2450N
U2	1	Power Supply, DC DC CONVERTER 5V 5W	SIP,11.6x7.5mm	Traco Power	TSR 1-2450E
U3, U4, U6, U7, U8, U9	6	Motordriver H-Bridge	SOPowerPAD-8	Texas Instruments	DRV8251ADDAR